

ICS344: Information Security

ICS 344 Course Project

Phase 2: Visual Analysis with a SIEM Dashboard

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Objectives:

This phase demonstrates security event monitoring using Splunk to:

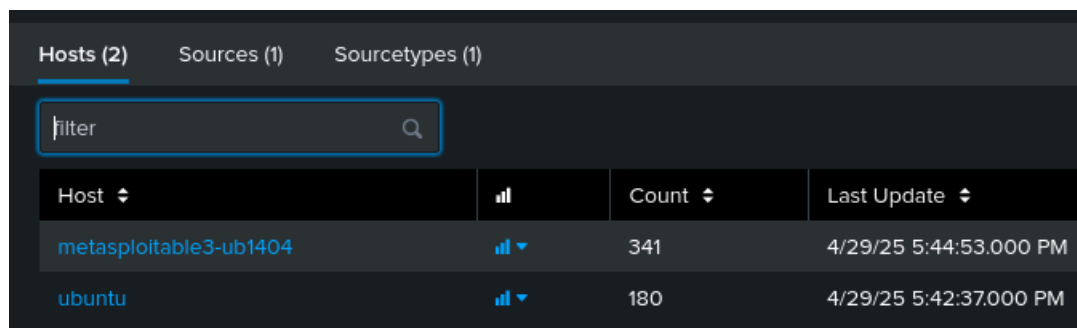
- Forward SSH authentication logs
- Analyze brute force patterns
- Visualize attack events

Requirements

- Splunk Enterprise
- Splunk Universal Forwarder v9.4.1
- SSH auth logs from Metasploitable3

```
Active forwards:
  10.0.2.12:9997
Configured but inactive forwards:
  None
vagrant@metasploitable3-ub1404:~$
```

This screenshot shows an active port forward (10.0.2.12:9997) on Metasploitable3, indicating an established connection. No inactive forwards are listed.



The screenshot shows the Splunk interface with a table of monitored hosts. The table has four columns: Host, a signal strength indicator, Count, and Last Update. Two hosts are listed: metasploitable3-ub1404 and ubuntu. The metasploitable3-ub1404 host has a higher log count (341) compared to the ubuntu host (180).

Hosts (2) Sources (1) Sourcetypes (1)			
filter			
Host		Count	Last Update
metasploitable3-ub1404	📶	341	4/29/25 5:44:53.000 PM
ubuntu	📶	180	4/29/25 5:42:37.000 PM

This image displays the two hosts that were monitored: Metasploitable3 and Ubuntu. The count of logs collected from each host is shown. Metasploitable3 has a significantly higher log count

(341 events), indicating more activity, which is expected given that it is the victim machine under attack. The logs from Ubuntu (180 events) are comparatively lower, indicating less activity during this analysis.

Data Summary			
Hosts (2)		Sources (1)	Sourcetypes (1)
filter			
Source ▾		Count ▾	Last Update ▾
/var/log/auth.log		521	4/29/25 5:44:53.000 PM

The data source for the logs is /var/log/auth.log, which contains authentication logs that are crucial for monitoring unauthorized login attempts. The total log count from this source is 521, which reflects both the victim and attacker interactions with the system.

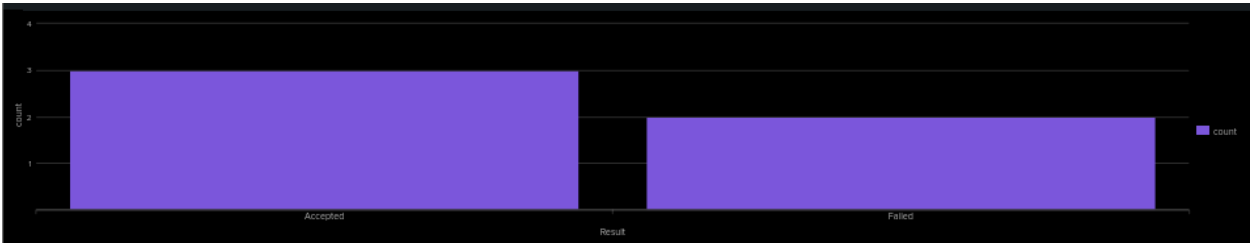
i	Time	Event
>	4/14/25 8:43:27.000 PM	Apr 14 17:43:27 metasploitable3-ub1404 sshd[2556]: Failed password for vagrant from 10.0.2.6 port 52550 ssh2 host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
>	4/14/25 8:39:08.000 PM	Apr 14 17:39:08 metasploitable3-ub1404 sshd[2512]: Failed password for invalid user msfadmin from 10.0.2.6 port 34394 ssh2 host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog

This image illustrates failed login attempts on Metasploitable3, specifically for the user 'vagrant' and 'msfadmin.' These failed attempts are from an attacker trying to gain unauthorized access

through SSH (port 22). This event is crucial for understanding the attacker's methods and timing during the attack.

i	Time	Event
>	4/14/25 8:20:06.000 PM	Apr 14 17:20:06 metasploitable3-ub1404 sshd[2307]: Accepted password for vagrant from 10.0.2.6 port 44163 ssh2 host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
>	1/8/25 1:03:35.000 PM	Jan 8 10:03:35 ubuntu sshd[1814]: Accepted password for vagrant from 10.0.2.2 port 61112 ssh2 host = ubuntu source = /var/log/auth.log sourcetype = syslog
>	10/29/24 10:26:05.000 PM	Oct 29 19:26:05 ubuntu sshd[962]: Accepted password for vagrant from 10.0.2.2 port 54757 ssh2 host = ubuntu source = /var/log/auth.log sourcetype = syslog

This image displays successful login attempts to Metasploitable3 and Ubuntu machines using the 'vagrant' account. These login attempts demonstrate successful authentication and potential compromise of the system, which will be useful for evaluating the attack's effectiveness.



The bar chart visualizes the results of login attempts on the victim machine. The 'Accepted' login attempts significantly outnumber the 'Failed' ones, suggesting that the attacker was able to successfully compromise the system during the attack. This result is consistent with the successful login events seen earlier.